Gómez's ΩH_q and the diameter-8 entries of the bipartite table

Doğa Özden 26 October 2026 Draft for Gabriela Araujo-Pardo and Nacho López; corrections very welcome

The entries. Six $D = 8$ entries of the bipartite table (Miller–Širáň 2013, Table 4, as on the CombinatoricsWiki, credited to Delorme–Gómez–Quisquater) are the orders of Gómez's compound graphs ΩH_q (J. Gómez, PhD thesis, UPC 1986, §2.9.3, values on p. 46). Here $B_1 = I(H(q))$ is the incidence graph of the split Cayley hexagon, $N_1 = 2(q^6 - 1)/(q - 1)$, $\Delta_1 = q + 1$, $D_1 = 6$, $q + 3 = 2\alpha + \beta$ with $\beta \in \{0, 1\}$, and $|\Omega H_q| = 5(\alpha + 1)N_1$:

cell	q	α	copies of $I(H(q))$	order
(10,8)	7	5	25	1,176,480
(11,8)	8	5	25	2,246,940
(12,8)	9	6	30	4,650,100
(14,8) and (15,8)	11	7	35	14,172,480
(16,8)	13	8	40	36,201,060

The thesis states $D \le D_1 + 2 = 8$ without proof (§2.8.2 says the proofs in that part are omitted). The cited paper, Delorme–Gómez–Quisquater, "On large bipartite graphs", we could not find in print.

The construction as defined. 5α copies of B_1, all carrying the same labels, and $5N_1$ new vertices (t, k), $(t, \bar{k})$, $k \in \mathbb{Z}_5$, t a label. Following Figs. 2.21a/b, each new vertex is joined to 2α of the 5α copies, and, by the convention of §2.8.9 (Fig. 2.18) that §2.9.3 refers to, a new vertex with label t is joined only to vertices labelled t (point t, or line $\sigma(t)$ for the pairing σ of point and line labels, which the thesis leaves open).

Observation. As defined, ΩH_q has diameter at least 9 for each q in the table.

1. *One label.* Suppose the new vertex x meets only copies of the vertex t (this is one reading of the figures). Since $2\alpha < 5\alpha$, x misses some copy j. Let v be opposite t in $H(q)$ (distance 6 in B_1). Project a path from x to the copy of v in copy j onto B_1 by labels: copy edges project to edges, a pass through a new vertex costs two edges and does not move the projection, and at least one pass is needed to change copy. So the path has at least $1 + 2 + 6 = 9$ edges.
2. *Two labels.* Under the other reading, x meets point t in some copies and line $\sigma(t)$ in others (σ arbitrary). Counting projected walks shows that, in a copy x misses, at most $A + B + 1$ of the q^5 elements opposite $\sigma(t)$ (or t) are within distance 7 of x, with $A + B + 1 < q^5$ for every $q \ge 7$ (e.g. 15,459 < 16,807 at $q = 7$, 39,763 < 59,049 at $q = 9$). So again some vertex is at distance ≥ 9.

Computer checks. All six graphs were rebuilt at the thesis orders by two separately written programs, from two separate constructions of $H(q)$, and in both readings of the figures. A breadth-first search from a new vertex gives eccentricity 9 in every case (all vertices at distance 9 lie in copies it misses, opposite its label). For ΩH_7 and ΩH_9 the exact diameter, 9, was computed by breadth-first search from one vertex in every orbit of an automorphism group whose generators were checked edge by edge. An explicit pair at distance 9 is recorded for every cell.

What we do not claim. Only that the construction as published does not reach diameter 8. We do *not* claim that no $(\Delta, 8)$ graph of these orders exists. Giving each copy its own labelling (which the thesis does not describe) reaches diameter 8 at the same order $5(\alpha + 1)N_1$ for $q = 2$ and $q = 3$; we found no such labelling for $q = 4, 5$, and have no argument either way for $q \ge 7$. Our own records at $(11, 8)$ and $(15, 8)$ (2,258,520 and 18,128,880, explicit Haar graphs) do not depend on any of this.

We would be grateful for your reading, in particular if we have misread the thesis's wiring or if you know of the Delorme–Gómez–Quisquater manuscript.

Files (https://github.com/dogaozden/degree-diameter, by branch):

- proofs and computations: `claude/own-12-8-cell-ig5jhd`, `research/squads/cells/d12D8_table.md`;
- independent re-derivation (own code): `claude/omega-dq-referee-bzzlid`, `research/referee/omega-dq.md`;
- a pair at distance 9 for every cell: `claude/table-audit-2-p7x6hn`, `research/projects/table-audit/checks/witness_omega_thesis.txt`.